

Deterministic pose-spline dynamics estimator

Estimator structure

pose-only continuous-time spline -> analytic derivatives
fixed-spline gradient matching -> shared physical parameters and lag
default physical initialization: exact nominal chart origin
previous estimator result loading: explicit opt-in only
shooting nodes: none
continuity constraints: none
sensor/IMU channels in loss: none

Initial estimate

source kind: nominal
source path: None
source mass [kg]: 2.35159759081
selected initial mass [kg]: 2.35159759081

Selected strict-ZOH parameters

mass [kg]: 2.76403114926
inertia [kg m²]:
[0.0730140972893, -0.00161358252189, 0.000176992247498]
[-0.00161358252189, 0.147719500677, 0.000148488837797]
[0.000176992247498, 0.000148488837797, 0.219768767228]
principal moments [kg m²]: [0.0729790402146, 0.147754046164, 0.219769278815]
inertia triangle margin [kg m²]: 0.000963807563077
CoG offset [m]: [-0.00273989441729, -0.00063359669019, -0.0164988909663]
rotor force effectiveness: [1.00385134748, 1.03699761142, 0.971521220562, 0.988781987866]
lag [s]: 0.042107553641
joint dynamics loss: 0.381803362452
soft-prior cost: 0.0472436555356

Smooth physical coordinates

log_mass_scale	0.161595252415
log_second_moment_cholesky_xx_scale	0.412903810128
log_second_moment_cholesky_yy_scale	0.0584040628752
log_second_moment_cholesky_zz_scale	0.00167164705019
normalized_second_moment_cholesky_yx_offset	0.0165470221322
normalized_second_moment_cholesky_zx_offset	-0.00517946083133
normalized_second_moment_cholesky_zy_offset	-0.00729098665536
cog_offset_x_m	-0.00273989441729
cog_offset_y_m	-0.00063359669019
cog_offset_z_m	-0.0164988909663
force_effectiveness_contrast_1	-0.0229708416954
force_effectiveness_contrast_2	0.039991149515
force_effectiveness_contrast_3	0.0130266485832

Spline and collocation

collocation step [s]: 0.01

Bag failure_1

selected spline knot spacing [s]: 0.1
dynamics loss: 0.665921339573
estimated forward position RMSE [m]: 22.9873796594
estimated + external wrench position RMSE [m]: 0.568534397352
external wrench improves estimated free rollout: True

nominal forward position RMSE [m]: 62.7634910707
estimated improves nominal: True
residual wrench mean: [0.10078 0.079549 0.053164 -0.030064 -0.010033 -0.009113]
residual wrench RMSE: [1.037731 0.740037 1.446431 0.129928 0.658027 0.431766]
residual wrench trend/s: [0.091623 0.036736 -0.568611 0.003763 -0.019791 0.023336]
residual wrench cumulative impulse: [0.620786 0.455288 0.300261 -0.178606 -0.060761
-0.054521]

Bag failure_2

selected spline knot spacing [s]: 0.2
dynamics loss: 0.0976853853312
estimated forward position RMSE [m]: 51.290471179
estimated + external wrench position RMSE [m]: 0.960966861938
external wrench improves estimated free rollout: True
nominal forward position RMSE [m]: 41.6709498739
estimated improves nominal: False
residual wrench mean: [0.081244 -0.362001 -0.021131 0.026764 0.026084 0.01426]
residual wrench RMSE: [0.212628 0.446652 0.710803 0.045669 0.247306 0.088943]
residual wrench trend/s: [-0.028776 0.029652 -0.281507 -0.010006 0.047393 0.003425]
residual wrench cumulative impulse: [0.448619 -1.965847 -0.125054 0.145996 0.137555
0.077043]

Smoothstep continuation

width=0.5: delay=0.0421280007s, cost=0.428607389311, nfev=19
width=0.2: delay=0.0422042122s, cost=0.428735805507, nfev=3
width=0.05: delay=0.0421075536s, cost=0.429026795419, nfev=5

Strict-ZOH polish

delay=0.0381075536s: screening=0.429624556151, refined=nan, selected=False
delay=0.0391075536s: screening=0.429264517727, refined=0.42926413224, selected=False
delay=0.0401075536s: screening=0.429190198154, refined=0.429189869865, selected=False
delay=0.0411075536s: screening=0.429312038073, refined=nan, selected=False
delay=0.0421075536s: screening=0.429047060054, refined=0.429047017988, selected=True
delay=0.0431075536s: screening=0.430112124046, refined=nan, selected=False
delay=0.0441075536s: screening=0.430150055274, refined=nan, selected=False
delay=0.0451075536s: screening=0.43020387885, refined=nan, selected=False
delay=0.0461075536s: screening=0.430379404523, refined=nan, selected=False